

Task Scheduling of Real-Time Traffic Information Processing Based on Digital Twins

Yang Liu¹, Ang Yang, Qingcheng Zeng, Yuhui Sun, Jing Gao², and Zhihan Lv³, *Senior Member, IEEE*

Abstract—The Intelligent Transportation System under Digital Twins can provide accurate data sources for traffic control. The present work focuses on the real-time information processing and task scheduling problems of the Internet of Vehicles (IoV) system based on Virtual Reality. They are the Quality/Distance Algorithm (QDA), Task Density Algorithm, Distance Balance Algorithm (DBA), and Bionic-DBA (B-DBA). The simulation experiment analysis suggests that the DBA algorithm takes the balance of travel distance into account and effectively improves task quality. The Utility Function in B-DBA and the Biological Heuristic Search Algorithm in Pareto Ant Colony Optimization play a critically important role in enhancing the overall task quality. In addition, a Transmission based on Privacy Protection (TPP) algorithm is designed to protect the attribute-based privacy information in the traffic information transmission system. This algorithm ensures that the real-time traffic information processing system resists various attacks from malicious nodes. It has been verified that when the number of selfish nodes accounts for 30%, the transmission efficiency of the TPP algorithm reaches 0.77. The research content has a practical reference value for providing users with continuous and high-quality IoV network services.

Index Terms—Digital twins, virtual reality, real-time traffic, information processing, task scheduling.

I. INTRODUCTION

AS THE urban population grows, so does the ratio of vehicles to population. Every metropolis faces the problem of effective management of traffic flow. With the increase of traffic problems, the modernization and digitization of traffic through intelligent systems has become a significant development trend [1], [2], [3]. China's Ministry of Transport of the People's Republic of China planned to promote the digital transportation system with the digital perception of transportation facilities, extensive coverage of information

networks, convenient and intelligent transportation services, online coordination of industrial governance, active technological application innovation, and robust network security by 2025 [4]. Today, the traditional transportation industry slowly transforms into an Intelligent Transportation System (ITS). In addition, Digital Twins (DTs) technology can revolutionize traffic management and operation mode. DTs refers to a digital version of a physical object or process constructed based on real-time two-way data exchange between digital and physical entities [5]. A traffic DTs system can be conceptualized as collecting traffic data in real-time from different physical systems, such as sensors, traffic signals, and traffic surveillance cameras, to create a network replica of the system. Although the concept of DTs replicates the idea of Cyber Physical Systems, the main challenge in constructing true DTs is to combine and link data from heterogeneous data sources of physical systems, to create network replicas of real-world traffic operations for real-time traffic management [6], [7], [8].

DTs technology can empower smart transportation in various aspects to meet future travel needs. DTs Traffic, also known as Virtual Traffic, incorporates the traffic data collected in real-time into the established traffic model system to realize virtual digital mapping of the traffic system. Chen *et al.* [9] generated and evaluated traffic optimization schemes through Big Data Analysis, Artificial Intelligence (AI), and traffic simulation techniques. In the DTs transportation field, video surveillance, Fused Millimeter-Wave Radar, and Holographic Perception of traffic elements, including motor vehicles, non-motor vehicles, and pedestrians, are used to build a mapping model of an entire traffic system in digital space for real-time analysis and tracking [10], [11], [12]. It can effectively solve traffic problems such as waste of traffic resources, rigid signal system functions, unpredictable traffic events, and rapid response.

The DTs transportation system based on Virtual Reality (VR) has made a further leap based on the Internet of Vehicle (IoV). The traffic management applications based on IoVs can assist the traffic management department in traffic diversion to prevent traffic congestion. Considering that Mobile Crowd-sensing is a new type of vehicle networking sensing mode, Lv *et al.* [13] and Chen *et al.* [14] made full use of the sensing, computing, and communication capabilities of ubiquitous smart devices to collect multi-dimensional information of physical traffic world. Task scheduling is a core issue in Mobile Crowd-sensing. Besides, designing an efficient

Manuscript received 5 May 2022; revised 2 July 2022; accepted 20 July 2022. Date of publication 18 August 2022; date of current version 1 November 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61902203 and in part by the Key Research and Development Plan—Major Scientific and Technological Innovation Projects of Shandong Province under Grant 2019JZZY020101. The Associate Editor for this article was H. Lv. (Yang Liu and Qingcheng Zeng are co-first authors.) (Corresponding author: Yang Liu.)

Yang Liu, Ang Yang, and Qingcheng Zeng are with the School of Shipping Economics and Management, Dalian Maritime University, Dalian, Liaoning 116026, China (e-mail: liuyang_87@dlmu.edu.cn; ang.yang@dlmu.edu.cn; qzeng@dlmu.edu.cn).

Yuhui Sun and Jing Gao are with the UniSA STEM, University of South Australia, Adelaide, SA 5001, Australia (e-mail: yuhui.sun@mymail.unisa.edu.au; mailjing@gmail.com).

Zhihan Lv is with the Department of Game Design, Faculty of Arts, Uppsala University, 75237 Uppsala, Sweden (e-mail: lvzhihan@gmail.com).

Digital Object Identifier 10.1109/TITS.2022.3196166

1558-0016 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

task scheduling algorithm is significant for improving task scheduling performance and reducing system operating costs.

Under the background of DTs technology, this research mainly focuses on the ITS of the VR-based IoVs. Besides, two key issues are analyzed combined with the mainstream data distribution technology and user application requirements: the real-time information processing method of IoVs and the intelligent perception task scheduling algorithm of traffic group to ensure accurate data sources for traffic control. The main contributions are: building a real-time traffic generation model, and designing a TPP algorithm based on secure multi-party computing theory. It improves the performance of traffic scheduling task allocation and ensures that the information processing system can resist various attacks from malicious nodes.

II. RELATED WORK

A. DTs Transportation System

From the perspective of digital traffic, traffic can be divided into real physical traffic and DTs traffic [15]. It is of great significance to expand the existing intelligent transportation platform into a DTs platform, use transportation models and AI to drive the virtual transportation world, and analyze and evaluate real transportation construction projects and improvement plans before they are implemented. Martínez-Gutiérrez *et al.* [16] proposed a new concept of DTs based on external services for the transportation system of Automatic Guided Vehicles to meet the material needs of collaborative industrial plants plan. The authors carried out simulation experiments in two different scenarios by defining an industrial Ethernet platform to verify the results of the DTs traffic system. With the new industrial revolution of digital transformation, manufacturing transportation processes can employ more intelligent and autonomous systems. Almeaibed *et al.* [17] adopted new safety designs to improve the resilience and safety of the entire independent vehicle system. They also put forward a standard DTs framework to facilitate data collection, processing, and analysis efficiencies to enable safe autonomous systems for intelligent manufacturing transportation end-to-end.

B. Smart Traffic Architecture Based on VR

Taking the transportation field as an example, VR technology has been applied to traffic safety education, rail transit construction, driving training, and in-station navigation. Rail transit simulation uses three-dimensional (3D) virtual and simulation technology to simulate the 3D environment of each stage and link of rail vehicles, including design, manufacture, operation, and maintenance [18]. Rozmus *et al.* [19] used 3D Scanning, Computer Simulation, and Instant VR technology to visualize the transportation environment, providing a reference for the area selection of traffic engineering. They proved that the method is effectively practical for transporting large goods. Lee *et al.* [20] completed the dynamic reconstruction of 2D planar images through a Model-Free Haptic Rendering Algorithm in an intelligent traffic environment, realizing the

user's real-time feeling of the transportation environment and the tactile interaction with the remote environment.

C. Task Scheduling in the IoV

IoVs have become an essential part of the information age. The IoV is a specific application of the IoT extending to intelligent transportation, known as the IoT for vehicles [21]. It receives information from and sends information to the car on the road through the sensors, onboard units, communication modules, and other devices installed on the vehicle. Information is transmitted to the data center through roadside units, base stations, and other equipment [22].

Feng *et al.* [23] reported that the management of the vehicle through the IoVs platform could visualize the running status of the car in real-time and monitor the safe state of cars; it could also check the location of the vehicle and perform remote dispatch and other management to improve the efficiency of enterprise management and improving economic benefits. Xu *et al.* [24] used AI based on the powerful real-time information collection capability of the IoVs to optimize edge services to solve practical task scheduling problems better.

According to the above research, the development of intelligent transportation has become even more diversified. However, there are few studies on the real-time processing of traffic information and task scheduling in the DTs scenario. This work will creatively discuss this research gap. The operation efficiency of the intelligent transportation system is promoted by constructing a real-time traffic generation model and proposing the corresponding task scheduling algorithm.

III. TRAFFIC INFORMATION PROCESSING AND TASK SCHEDULING IN DTs SCENARIOS

A. DTs Construction in Smart Transportation Scenarios

The evolution of computing technology represented by Cloud Computing, Big Data, and AI, as well as the leap of Tandem Relay Technique represented by All-Optical Networks, 5G/6G, and Wi-Fi 6, makes people put forward higher expectations for digital technology. The DTs platform collects real-time images in an actual road monitoring system through the road test camera, detects and senses various targets (vehicles, driving, etc.), and then extracts the target data. After modeling, the data is synchronously integrated into the DTs environment [25], [26], [27]. The digitization of infrastructure and multi-source heterogeneous physical perception is the base in the actual construction of ITSSs. The data is uploaded to the computing platform with twin functions for modeling, reconstruction, and accurate perception using various low-latency network combinations as channels. Thus, the structured digital data is applied to multiple traffic businesses to realize the functions of traffic state deduction and automatic analysis and prediction via Big Data.

The 3D model collected based on the high-precision map at the perception layer digitizes the road network, roads, signs, markings, buildings, poles, etc., and matches the corresponding feature information. The data layer distributes, stores, and manages the data collected in different ways according

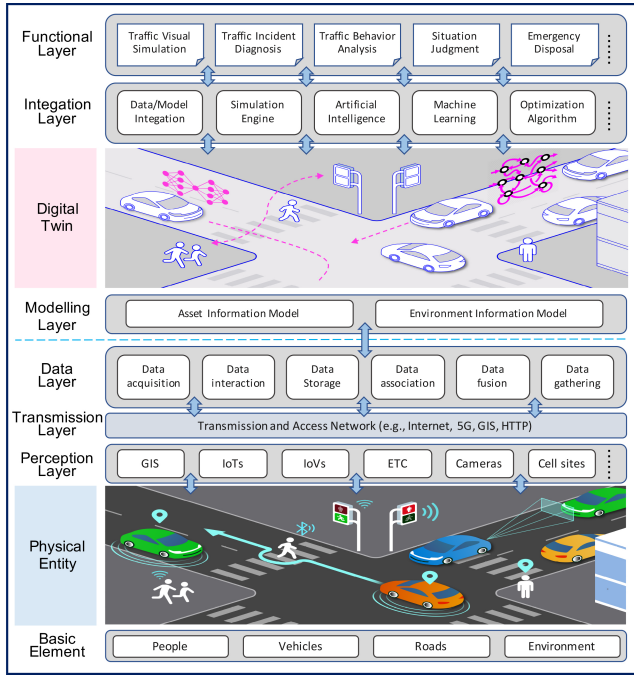


Fig. 1. DTs architecture in smart transportation scenarios.

to business needs [28]. The computing layer uses computing resources to fuse and accurately match the life cycle data of things with the dynamically collected data, convert physical quantities into machine-recognizable language, and build organic DTs. The functional layer is oriented to the business system. It implements the twin reproduction of traffic scenes, the target recognition, tracking and restoration of traffic participants, and intelligent analysis of target behavior and traffic status based on automatic recognition and processing of machine language. This layer can also conduct a situation analysis and traffic event prediction according to the target. Fig. 1 presents the DTs architecture in smart transportation scenarios.

The traffic scene includes four elements: people, vehicles, roads, and the environment. In the construction of DTs, in addition to modeling a single element, it is also necessary to construct its influence on different aspects and respond to the impact of other factors on the DTs in time.

In innovative traffic scenarios, DTs technology emphasizes active operation, i.e., managing the entire expressway. Data transmission can be achieved through Electronic Toll Collection, cameras, IoVs terminals, roadside units, cellular base stations, and even satellites. Cloud or On-Site Edge Computing Nodes can calculate and process data. The expressway management department can quickly realize active safety and fine-grained control of all elements of the expressway based on the perceived traffic flow, road surface, weather, and accident data. For drivers, the traffic DTs technology has a precise perception capability with an accuracy rate of 95% and a communication capability with an end-to-end delay of 300ms. Based on the traffic DTs technology, the system can also provide accompanying Location Based Services for car owners and push the services to car owners through WeChat, map App, etc., to improve the driving experience [29], [30].

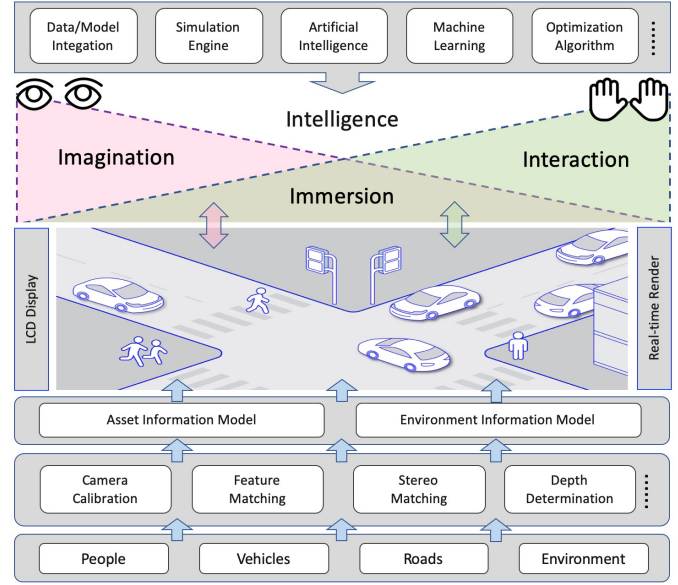


Fig. 2. Architecture of the VR-based DTs intelligent transportation system.

B. Real-Time Traffic Generation Model Based on Virtual Online Simulation

Figure 2 shows the VR-based architecture of DTs intelligent transportation system. The basis of the vehicle in the simulation system is primarily the traffic flow information collected in the natural traffic scene. Converting the relevant data can restore the traffic situation at the current moment in real-time. The road network construction tool configures the regional road network and adapts and transforms the road network of other data formats. Traffic flow data is increasingly available in video, lidar, and Global Position System (GPS) sensors. These technologies can reconstruct spatiotemporal data acquired from existing road traffic flow sensors, synthesize new traffic flows from limited sample trajectories, and learn behavioral patterns and independent features from traffic monitoring datasets to generate traffic flow [31].

The GPS terminal on the vehicle is used to measure the data generated by the road sections with significant traffic in the traffic scene to accurately understand the characteristics of urban traffic flow [32], [33], [34]. The entire traffic network is detected as comprehensively as possible by deploying stations at critical points of the urban road network. Here, N pairs of i, j are set in the traffic road network. The actual travel demand $\hat{d}_{i,j}$ and the estimated travel demand $d_{i,j}$ of the i, j pair can be expressed as:

$$\sum_i \sum_j P_{i,j}^{(r)} d_{i,j} = v^r \left(r \in \hat{L}; i, j \in N \right) \quad (1)$$

$$\sum_i \sum_j P_{i,j}^{(r)} \hat{d}_{i,j} = v^r \left(r \in \hat{L}; i, j \in N \right) \quad (2)$$

The error between actual travel demand and estimated travel demand is calculated according to Eq. (3).

$$\varepsilon = \sum_i \sum_j P_{i,j}^{(r)} (\hat{d}_{i,j} - d_{i,j}) \left(r \in \hat{L}; i, j \in N \right) \quad (3)$$

Assume that:

$$Z_{i,j} = \dot{d}_{i,j} - d_{i,j} \quad (4)$$

then, according to the Maximum Possible Relative Error Algorithm, there is:

$$\varepsilon = \sum_i \sum_j P_{i,j}^{(r)} Z_{i,j} = 0 \quad (5)$$

If the station cannot observe the flow of any road section between the pair of i, j , then $P_{i,j}^{(r)} = 0$. At this time, the value of the error $Z_{i,j}$ can be infinite. Let D represent the set $d_{i,j} = 0$. Then, according to Eq. (5), there are:

$$Z_{i',j'}/d_{i',j'} = - \sum_{i,j \neq i',j'} \sum_{p_{i,j} Z_{i,j}/p_{i',j'} d_{i',j'} = 0} \quad (6)$$

$$Z_{i',j'}/d_{i',j'} = V_{i,j} \quad (7)$$

The following two equations can be derived from Eq. (6) and Eq. (7):

$$V_{i',j'} = - \left(\sum_{i,j \neq i',j'} \sum_{p_{i,j}^{(r)} Z_{i,j}} + \sum_{i,j \neq i',j'} \sum_{p_{i',j'}^{(r)} d_{i',j'}} \right) / p_{i',j'}^{(r)} d_{i',j'} \quad (8)$$

$$V_{i,j} = (\dot{d}_{i,j} - d_{i,j}) / d_{i,j} = \dot{d}_{i,j} / d_{i,j} - 1 \geq -1 \quad (9)$$

If:

$$C = \sum_{i,j \neq i',j'} \sum_{p_{i,j}^{(r)} Z_{i,j}} / p_{i',j'}^{(r)} d_{i',j'} \quad (10)$$

then:

$$V_{i',j'} \leq C + v^r / p_{i',j'}^{(r)} d_{i',j'} - 1 \quad (11)$$

The more significant the proportion of the observed flow in the total flow of a particular section, the smaller the relative error $V_{i',j'}$.

The operation basis of the online traffic simulation system is to adapt to the actual traffic network conditions. The traffic data observation points extract traffic distribution information in the road network. Assume that the time interval between the two collections of the data observation site is Δt . Then, at the t_n moment, the number of vehicles observed by an edge site is d_{t_n} , the set is v , the speed is v , and the acceleration is $\ddot{x}$. From this, the vehicle generation model at the edge observation site in the simulation system can be described as:

$$\begin{aligned} & \{v'_{vehicle-count+i} \ (i = 0, 1, \dots, d_{t_n})\} \\ & = \{v_i \ (i = 0, 1, \dots, d_{t_n})\} \end{aligned} \quad (12)$$

$$\begin{aligned} & \{\dot{x}'_{vehicle-count+i} \ (i = 0, 1, \dots, d_{t_n})\} \\ & = \{\dot{x}_i \ (i = 0, 1, \dots, d_{t_n})\} \end{aligned} \quad (13)$$

$$\begin{aligned} & \{\ddot{x}'_{vehicle-count+i} \ (i = 0, 1, \dots, d_{t_n})\} \\ & = \{\ddot{x}_i \ (i = 0, 1, \dots, d_{t_n})\} \end{aligned} \quad (14)$$

Real road network traffic information comes from many aspects. At the t_n moment, the observation station D collects the information about the vehicle $d_{t_n}^{(D)}$. Meanwhile, the observation station $\{A, B, C\}$ collects the data of vehicles

$\{d_{t_n}^{(A)}, d_{t_n}^{(B)}, d_{t_n}^{(C)}\}$, respectively. At the t_{n-1} moment, the vehicle information in the simulation system at the observation station $\{A, B, C\}$ can be written as Eq. (15).

$$(Vehicle - Set)_{t_{n-1}} \times \cup \left\{ \sum v'_{vehicle-count+i} \ (i = 0, 1, \dots, d_{t_n}^{(A,B,C)}) \right\} \quad (15)$$

After the elapsed Δt time, the number of vehicles in the $\{A, B, C, D\}$ area in the system can be expressed as:

$$\begin{aligned} (Vehicle - Count)_{t_n} &= (Vehicle - Count)_{t_{n-1}} \\ &- \sum_{i \in (A,B,C), j=D} \gamma_{i,j} p_{i,j}^{(i,j)} d_{t_{n-1}}^{(A,B,C)} \end{aligned} \quad (16)$$

$$\gamma = \frac{count \{v''\}}{p_{i,D}^{(i,D)} d_t^{(i)}} \quad (17)$$

$$v'' = \{v' \in (position(v) \notin area(A, B, C, D))\} \quad (18)$$

Introducing the error coefficient μ existing in the simulation process can identify the number of vehicles generated at the observation site D , as shown in Eq. (9).

$$d_{t_n}^{(D)} - (count \{v''\}) \mu \quad (19)$$

The final model generation process of the entire real-time traffic observation point can be regarded as a recursive process from edge points to interior points.

C. Real-Time Information Processing of IoVs Via Collaborative Perception

Various computer vision detection, recognition, tracking, and segmentation algorithms based on video can accurately identify the road (pavement, markings, cracks, potholes, roadbed, building facilities, guardrails, etc.), road participants (people, vehicles, non-motor vehicles, etc.), road environment (water, snow, fog, etc.), and road abnormal events. They have the incomparable advantages of other sensors [35], [36].

The collaborative control of vehicles at traffic intersections discussed here is vital for realizing multi-vehicle collaborative decision-making and ensuring vehicle safety at the corner. Cellular Vehicle Network (C-V2X) supports longer communication distance, better non-line-of-sight performance, more substantial reliability, higher capacity, and better congestion control than traditional networks. Therefore, cooperative vehicle control at traffic intersections is proposed based on C-V2X communication. Fig. 3 displays the prominent architecture. It can be seen that the On Board Unit (OBU), Road Side Unit (RSU), and the communication equipment together constitute the cooperative vehicle control architecture. RSU mainly collects the real-time status of vehicles in different lanes; OBU is responsible for sending the vehicle's data to the RSU to guide the vehicle to drive safely. Fig. 4 is the schematic diagram of the vehicle collision area at the traffic intersection.

Take Fleet 1 going straight and Fleet 2 turning left as an example for analysis. The area where the two teams collide is A_{ABCD} . When Fleet 2 turns left, the length from the

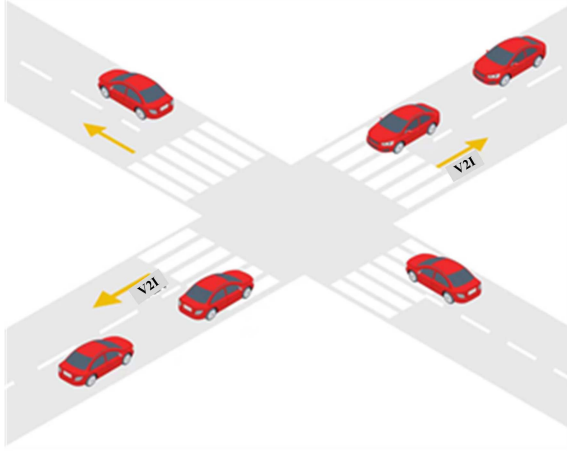


Fig. 3. Cooperative vehicle control at traffic intersection based on C-V2X communication.

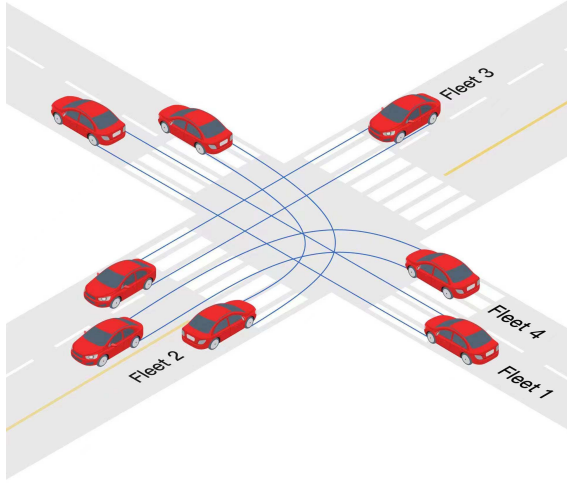


Fig. 4. Vehicle collision area at traffic intersection.

intersection to the collision area is calculated according to Eq. (20).

$$L_{R1} = \sqrt{\left(R_{left} - \frac{1}{2}L_s\right)^2 - \left(\frac{1}{4}L_{crossing} + \frac{1}{2}L_s\right)^2} \quad (20)$$

In Eq. (20), L_s represents the body width.

The time T_{PZ1} of Fleet 1 to reach the collision area at the current speed v_1 can be expressed as Eq. (21).

$$T_{PZ1} = \frac{L_1 + L_{R1}}{v_1} \quad (21)$$

In Eq. (20), L_1 refers to the distance from Fleet 1 to the intersection.

The deflection angle θ_{PZ2} and time T_{PZ2} for Fleet 2 to reach the collision area at the current speed v_2 can be written as:

$$\theta_{PZ2} = \arcsin \frac{\frac{1}{4}L_{crossing} - \frac{1}{2}L_s}{R_{left}} \quad (22)$$

$$T_{PZ2} = \frac{L_2 + \frac{2\pi R_{left}}{360^\circ} \theta_{PZ2}}{v_2} \quad (23)$$

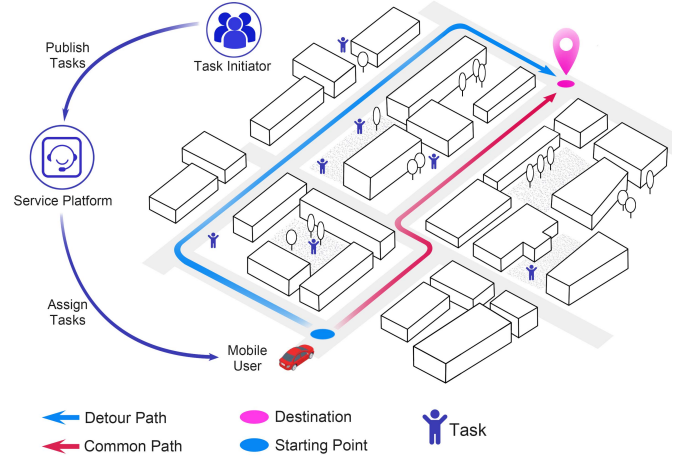


Fig. 5. Distance-constrained mobile crowd-sensing system.

where L_2 denotes the distance of Fleet 2 from the intersection.

The time T_{LK1} and T_{LK2} of Fleets 1 and 2 to leave the collision area can be expressed as:

$$T_{LK1} = \frac{R_{left} + \frac{1}{2}L_s + L_1 + S_1}{v_1} \quad (24)$$

$$T_{LK2} = \frac{L_2 + \frac{2\pi R_{left}}{360^\circ} \theta_{LK2}}{v_2} \quad (25)$$

D. Traffic Crowd-Sensing Task Scheduling Algorithm

In actual traffic scenarios, mobile users and sensing tasks in a Mobile Crowd-sensing System are usually distributed in different locations in the area. Therefore, users need to consider their travel distance budget to move to a specific target location. The Crowd-sensing System is composed of three parts: service platforms, task initiators, and mobile users $U = \{u_1, u_2, \dots, u_n\}$. The task initiator issues a series of dynamic tasks obeying the Poisson distribution. The task quality function $A_{p_j}(K)$ depends on how many times the job is completed. When K is an odd number, there are:

$$A_{p_j}(K) = \sum_{r=\frac{K+1}{2}}^K \binom{K}{r} p_j^r (1-p_j)^{K-r} \quad (26)$$

Other cases besides this is:

$$A_{p_j}(K) = \sum_{r=\frac{K}{2}+1}^K \binom{K}{r} p_j^r (1-p_j)^{K-r} \quad (27)$$

$A_{p_j}(K)$ is a non-decreasing function of K . Among them, K represents the execution times of the task t_j , and p_j stands for a task attribute.

If the arrivals of mobile users follow a Poisson Distribution, each user has an initial location and a destination. A user will complete the registration of personal information after arrival. In the Mobile Crowd-sensing System of Fig. 5, each user u_i has a travel distance B_i ; T_i represents the set of all tasks that are not expired upon u_i 's arrival.

This paper proposes four Polynomial Heuristic Algorithms: the Quality/Distance Algorithm (QDA), the Task Density Algorithm (TDA), Distance Balance Algorithm (DBA), and Bionic-DBA, (B-DBA). In the implementation process of QPA, the core idea is to achieve the local optimum and select high-quality and low-cost tasks. The utility function constructed accordingly can be expressed as Eq. (28).

$$Utility(t_j) = \frac{\Delta F_{p_j}}{dist(tail, j)} \quad (28)$$

In Eq. (28), t_j represents a candidate task, and $dist(tail, j)$ denotes the travel distance from the end of the current path to the task location.

QPA compares the utility values of different candidate tasks and selects the task satisfying the distance constraint with the maximum utility value is chosen as the next task. TDA has a higher task density than QPA to guide the user to the task-intensive area to obtain more gains, improving the local searchability. To achieve this, a new utility function is defined as:

$$Utility(t_j) = \frac{Weighted(\Delta F_{p_j})}{dist(tail, j)} \quad (29)$$

$$Weighted(\Delta F_{p_j}) = \frac{1}{\sqrt{2\pi}\sigma} \sum_{k=1}^{T_i} \left(\Delta F_{ik} \cdot \exp\left(-\frac{d_j^2 k}{2\sigma^2}\right) \right) \quad (30)$$

where σ stands for the standard deviation describing the radius of the task's impact area.

QPA nor TDA considers the effect of task selection on the distance balance. Therefore, DBA is put forward here. The utility function used to measure the benefits of a candidate task can be expressed as Eq. (31).

$$Utility(t_j) = \frac{\Delta F_{p_j} \cdot (B_i - length(P_i) - dist(tail, j) - e_{i,j})^\alpha}{dist(tail, j)^\gamma} \quad (31)$$

In Eq. (31), α represents the weight parameter of the distance balance, and γ signifies the distance weight between the end task position and the candidate task position.

On this basis, to solve the Crowd-sensing task scheduling problem, the Pareto Ant Colony (PAC) algorithm is chosen as the Object Heuristic Search algorithm, which is integrated with the DBA algorithm. The preference function of the PAC algorithm can be written as Eq. (32).

$$Pre(t_j) = \tau_{tail,j}^\epsilon \cdot \left(\frac{\Delta F_{p_j}}{dist(tail, j)} \right)^\beta \quad (32)$$

In Eq. (32), $\tau_{tail,j}$ represents the local pheromone from the last task to the task t_j ; ϵ and β are weight parameters.

After all the ants have decided their respective paths, the two ants with the highest sum of task quality increase are selected, and their corresponding global information elements increase by τ_0 and $\tau_0/2$.

E. Privacy Protection Scheme for Real-Time Traffic Information Processing

Vehicles need to upload accurate event reports based on their perception to enable the traffic management server to respond to all kinds of emergencies quickly. This study proposes a message transmission strategy based on privacy protection, named Transmission based on Privacy Protection (TPP). Under this transmission strategy, the real-time traffic information processing system can resist various attack strategies of malicious nodes. In addition, the attribute-based encryption algorithm is combined with the message transmission process to protect the user's attribute privacy.

The message processing module uses the Advanced Encryption Standard (AES) symmetric encryption algorithm to encrypt the message after it is generated and before it is sent. The AES encryption routine begins by copying the 16-byte input array into a 4×4 -byte matrix called State. Next, an attribute-based encryption algorithm is used to encrypt the encryption algorithm's key. Then, the node sends the encrypted key to the local cloud. Finally, the key is encrypted again by the message processing module. The result is returned to the node; the node sends the encrypted message and key to the next-hop node. This algorithm uses a random b to define the transfer key. b is only visible to the corresponding node.

Under the TPP strategy, nodes can obtain an appropriate amount of virtual currency by transmitting messages. If any node in the ensemble G_p discards messages, the report message m_r will not reach the local cloud c_k , and all nodes in the ensemble G_p will not be rewarded. Therefore, nodes will try to cooperate with the message transmission process as much as possible to obtain sufficient virtual currency to send locally generated messages.

F. Simulation Experiment Design

The simulation experiment aims to evaluate the performance of the task scheduling algorithm proposed here. The algorithm is implemented by JAVA programming, and the Euler distance is used as the travel distance. All tasks are uniformly distributed in space; their arrival times and durations follow Poisson and normal distributions. Two algorithms, Optimal Transmission Allocation (OTA) and Pareto Ant Colony Optimization (PACO) are selected as comparisons to evaluate the performance of the four task scheduling algorithms reported above.

ONE simulation software is adopted for simulation to discuss the effectiveness of TPP privacy protection policy. The datasets used are INFOCOM06 and SIGCOMM09. The K-means algorithm is used to calculate the location information of the local cloud. The transmission range is 10 meters; the transmission speed is 250KB. The experiment is run 50 times to obtain the average result. The TPP scheme proposed here is compared with three other representative methods: Prophet Routing Algorithm, Copy Adjustable Incentive Scheme (CAIS) algorithm, and Trust Routing based on Social Similarity (TRSS) algorithm.

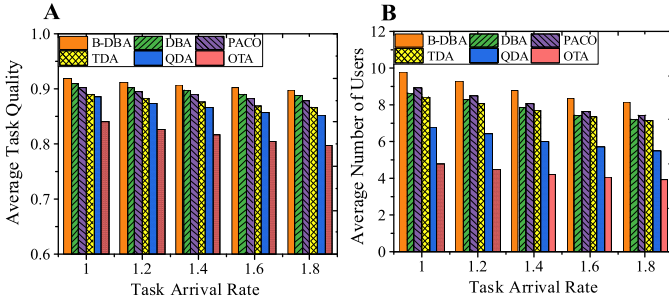


Fig. 6. Influence of task arrival rate on algorithm performance (A. average task quality; B. average number of users completing each task).

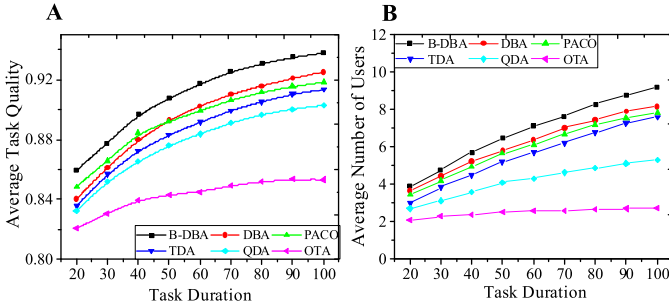


Fig. 7. Effect of task duration on algorithm performance (A. average task quality; B. average number of users completing each task).

IV. RESULTS AND DISCUSSION

A. Performance Evaluation of Task Scheduling Algorithms

The performance of several proposed task scheduling algorithms is compared in the actual scenario. Fig. 6 reveals the performance and variation trend of each algorithm after changing the task arrival rate. It can be seen that the task quality of each task scheduling algorithm and the number of users completing each task decrease accordingly with the growth of the task arrival rate. The algorithm performance is ranked in descending order of task quality: B-DBA, DBA, PACO, TDA, QDA, and OTA. This result indicates that DBA takes the balance of travel distance into account, improving the quality of tasks. The utility function in B-DBA and the Biological Heuristic Search Algorithm in the PACO algorithm significantly enhance the overall task quality. Fig. 7 provides each algorithm's performance and changing trend after changing the task duration. As the task duration increases, the task quality of each task scheduling algorithm and the number of users completing each task also increase. Except when the task duration is less than 40s, the performance of DBA is slightly weaker than that of the PACO algorithm. In other cases, the task quality performance of B-DBA and DBA is better than the other algorithms.

B. Evaluation of Traffic Information Processing and Privacy Protection Schemes

Fig. 8 to Fig. 10 respectively presents the transmission efficiency, transmission delay, and transmission cost of several data transmission algorithms under changing message life cycle. The results suggest that in the two datasets, the average transmission efficiency of the TPP algorithm is lower than that of the CAIS algorithm but significantly better than the Prophet Routing Algorithm and the TRSS algorithm. Accordingly, the

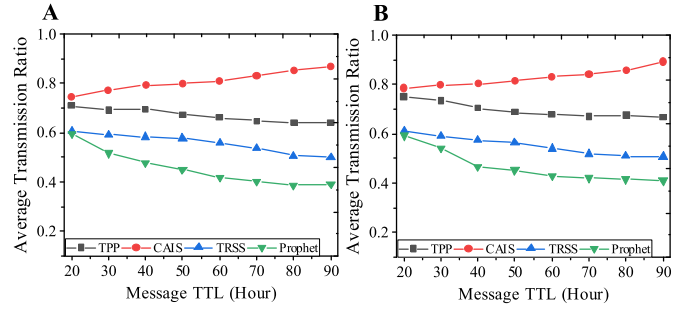


Fig. 8. Comparison of the average transmission efficiency of the algorithms under varying message life cycle (A. INFOCOM06; B. SIGCOMM09).

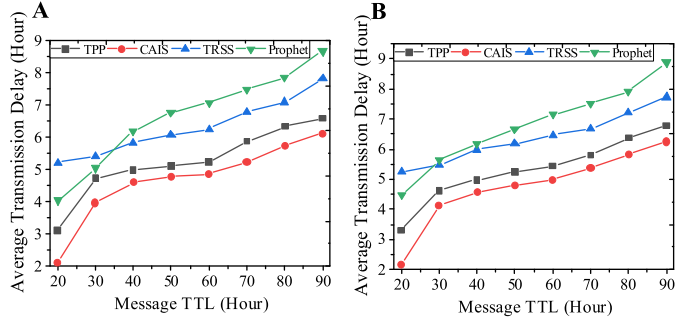


Fig. 9. Comparison of the average transmission delay of the algorithms under changing message life cycle (A. INFOCOM06; B. SIGCOMM09).

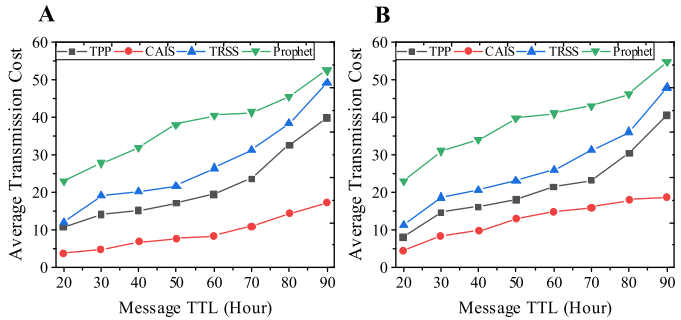


Fig. 10. Comparison of the average transmission cost of the algorithms under varying message life cycle (A. INFOCOM06; B. SIGCOMM09).

transmission efficiency decreases. With the extension of the message life cycle, the average transmission delay of various data transmission algorithms increases gradually. In the TPP algorithm, the message cannot be sent if the local cloud does not encrypt the transfer key. There is a slight rise in transmission delay compared with the CAIS algorithm. However, compared with Prophet Routing Algorithm and TRSS algorithm, the transmission delay of the TPP algorithm decreases by 35% and 20%, respectively. When the life cycle of the news is 40 hours, the transmission efficiency of the CAIS algorithm is 5% higher than that of the TPP algorithm.

Furthermore, the transmission efficiency, transmission delay, and transmission cost of different data transmission algorithms are compared under the indefinite quantity of selfish nodes. The specific results are shown in Fig. 11 ~ Fig. 13. The TPP algorithm is better than the TRSS algorithm and Prophet Routing Algorithm in terms of average transmission efficiency. When the number of selfish nodes accounts for 30%, the transmission efficiency of the TPP algorithm attains 0.77, about 25% higher than that of the TRSS algorithm and the Prophet Routing Algorithm, but 7% lower than the CAIS

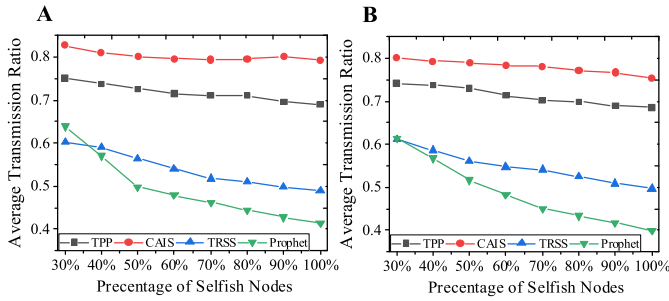


Fig. 11. Comparison of the average transmission efficiency of the algorithms when the number of selfish nodes changes (A. INFOCOM06; B. SIGCOMM09).

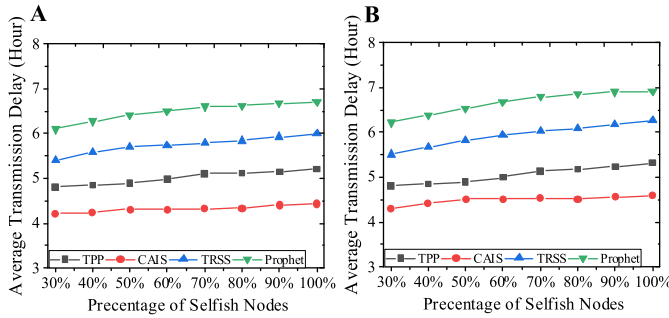


Fig. 12. Comparison of the average transmission delay of the algorithms.

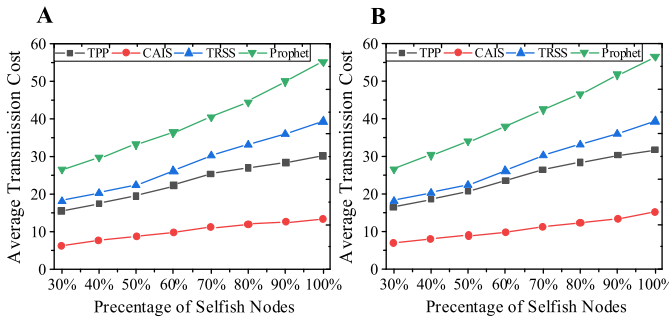


Fig. 13. Comparison of the average transmission cost of the algorithms when the number of selfish nodes changes (A. INFOCOM06; B. SIGCOMM09).

algorithm. Since the TRSS algorithm and Prophet Routing Algorithm cannot motivate selfish nodes to transmit messages, they are lower than the other three algorithms in terms of delay. When the selfish nodes account for 50%, the average transmission delay and transmission cost of the TPP algorithm is about 15% lower than those of the TRSS algorithm.

To resist different types of attacks, this section applies the security based mobile prediction algorithm and attribute-based encryption algorithm to the routing decision-making and message transmission stages. This only defines the fine-grained access control mechanism, but also protects the user's personal privacy information. Based on two real datasets, the experimental results show that the proposed algorithm has higher transmission efficiency and lower transmission delay compared with other security-based algorithms. Compared with non-security algorithms, there is only less performance degradation. The data distribution mechanism based on privacy protection provided here provides a new solution for the secure transmission of messages in the IoT.

V. CONCLUSION

The IoT, Cloud Computing, Internet, AI, Automatic Control, Mobile Internet, and other technologies are fully utilized in the transportation field. This paper focuses on the VR-based DTs transportation system based on this realistic background. A C-V2X communication-based collaborative vehicle control at a traffic intersection is implemented for the DTs traffic system. Besides, four Polynomial Heuristic algorithms are proposed for task scheduling problems. A TPP algorithm is designed based on the secure multi-party computation theory to protect the attribute-based privacy information in the traffic information transmission system. This algorithm shows excellent performance on both the INFOCOM06 and SIGCOMM09 datasets. The content discussed here is of great value for advancing DTs transportation systems and dealing with real traffic problems. However, in many swarm intelligence perception scenarios, the perception data show strong correlation in the time dimension and spatial dimension. This factor is not considered in the task scheduling problem here, so there are certain limitations. This problem will be discussed in detail in the follow-up research.

REFERENCES

- [1] M. Elsis, M. Q. Tran, K. Mahmoud, and D. E. A. Mansour, "Towards secured online monitoring for digitalized GIS against cyber-attacks based on IoT and machine learning," *IEEE Access*, vol. 9, pp. 78415–78427, 2021.
- [2] Q. Yuan, Y. Peng, X. Xu, and X. Wang, "Key points of investigation and analysis on traffic accidents involving intelligent vehicles," *Transp. Saf. Environ.*, vol. 3, no. 4, Oct. 2021, Art. no. tdab020.
- [3] N. R. Yusupbekov and V. B. Tarassov, "Integrated intelligent systems for industrial automation: The challenges of industry 4.0, information granulation and understanding agents," *Chem. Technol., Control, Manag.*, vol. 2018, no. 4, pp. 62–75, 2018.
- [4] W. Liu, S. Wang, Y. Lin, D. Xie, and J. Zhang, "Effect of intelligent logistics policy on shareholder value: Evidence from Chinese logistics companies," *Transp. Res. E, Logistics Transp. Rev.*, vol. 137, May 2020, Art. no. 101928.
- [5] Y. Lu, C. Liu, K. I.-K. Wang, H. Huang, and X. Xu, "Digital twin-driven smart manufacturing: Connotation, reference model, applications and research issues," *Robot. Comput. Integr. Manuf.*, vol. 61, Feb. 2020, Art. no. 101837.
- [6] K. Ding, F. T. S. Chan, X. Zhang, G. Zhou, and F. Zhang, "Defining a digital twin-based cyber-physical production system for autonomous manufacturing in smart shop floors," *Int. J. Prod. Res.*, vol. 57, no. 20, pp. 6315–6334, 2019.
- [7] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108952–108971, 2020.
- [8] P. Bellavista, C. Giannelli, M. Mamei, M. Mendula, and M. Picone, "Application-driven network-aware digital twin management in industrial edge environments," *IEEE Trans. Ind. Informat.*, vol. 17, no. 11, pp. 7791–7801, Nov. 2021.
- [9] D. Chen, P. Wawrzynski, and Z. Lv, "Cyber security in smart cities: A review of deep learning-based applications and case studies," *Sustain. Cities Soc.*, vol. 66, Mar. 2021, Art. no. 102655.
- [10] F.-Y. Wang, Y. Li, W. Zhang, G. Bennett, and N. Chen, "Digital twin and parallel intelligence based on location and transportation: A vision for new synergy between the IEEE CRFID and ITSS in cyberphysical social systems [society news]," *IEEE Intell. Transp. Syst. Mag.*, vol. 13, no. 1, pp. 249–252, 2021.
- [11] C. Qian, X. Liu, C. Ripley, M. Qian, F. Liang, and W. Yu, "Digital twin—Cyber replica of physical things: Architecture, applications and future research directions," *Future Internet*, vol. 14, no. 2, p. 64, Feb. 2022.
- [12] Y. H. Pan, N. Q. Wu, T. Qu, P. Z. Li, K. Zhang, and H. F. Guo, "Digital-twin-driven production logistics synchronization system for vehicle routing problems with pick-up and delivery in industrial park," *Int. J. Comput. Integr. Manuf.*, vol. 34, nos. 7–8, pp. 814–828, Aug. 2021.

- [13] Z. Lv, D. Chen, R. Lou, and Q. Wang, "Intelligent edge computing based on machine learning for smart city," *Future Gener. Comput. Syst.*, vol. 115, pp. 90–99, Feb. 2021.
 - [14] Z. Chen, C. Fiandrino, and B. Kantarci, "On blockchain integration into mobile crowdsensing via smart embedded devices: A comprehensive survey," *J. Syst. Archit.*, vol. 115, May 2021, Art. no. 102011.
 - [15] M. Gonzalez, O. Salgado, J. Croes, B. Pluymers, and W. Desmet, "A digital twin for operational evaluation of vertical transportation systems," *IEEE Access*, vol. 8, pp. 114389–114400, 2020.
 - [16] A. Martínez-Gutiérrez, J. Díez-González, R. Ferrero-Guillén, P. Verde, R. Álvarez, and H. Perez, "Digital twin for automatic transportation in industry 4.0," *Sensors*, vol. 21, no. 10, p. 3344, May 2021.
 - [17] S. Almeaibed, S. Al-Rubaye, A. Tsourdos, and N. P. Avdelidis, "Digital twin analysis to promote safety and security in autonomous vehicles," *IEEE Commun. Standards Mag.*, vol. 5, no. 1, pp. 40–46, Mar. 2021.
 - [18] Z. Yang, D. Xiang, and Y. Cheng, "VR panoramic technology in urban rail transit vehicle engineering simulation system," *IEEE Access*, vol. 8, pp. 140673–140681, 2020.
 - [19] M. Rozmus *et al.*, "Application of 3D scanning, computer simulations and virtual reality in the redesigning process of selected areas of underground transportation routes in coal mining industry," *Energies*, vol. 14, no. 9, p. 2589, May 2021.
 - [20] J. Lee, X. Zhang, C. H. Park, and M. J. Kim, "Real-time teleoperation of magnetic force-driven microrobots with 3D haptic force feedback for micro-navigation and micro-transportation," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 1769–1776, Apr. 2021.
 - [21] D. M. Manias and A. Shami, "Making a case for federated learning in the internet of vehicles and intelligent transportation systems," *IEEE Netw.*, vol. 35, no. 3, pp. 88–94, May 2021.
 - [22] P. Bagga, A. K. Das, M. Wazid, J. J. P. C. Rodrigues, K.-K.-R. Choo, and Y. Park, "On the design of mutual authentication and key agreement protocol in internet of vehicles-enabled intelligent transportation system," *IEEE Trans. Veh. Technol.*, vol. 70, no. 2, pp. 1736–1751, Feb. 2021.
 - [23] J. Feng, Z. Liu, C. Wu, and Y. Ji, "Mobile edge computing for the internet of vehicles: Offloading framework and job scheduling," *IEEE Veh. Technol. Mag.*, vol. 14, no. 1, pp. 28–36, Mar. 2019.
 - [24] X. Xu, H. Li, W. Xu, Z. Liu, L. Yao, and F. Dai, "Artificial intelligence for edge service optimization in internet of vehicles: A survey," *Tsinghua Sci. Technol.*, vol. 27, no. 2, pp. 270–287, Apr. 2022.
 - [25] T. Deng, K. Zhang, and Z.-J. Shen, "A systematic review of a digital twin city: A new pattern of urban governance toward smart cities," *J. Manag. Sci. Eng.*, vol. 6, no. 2, pp. 125–134, Jun. 2021.
 - [26] A. J. Saroj, S. Roy, A. Guin, and M. Hunter, "Development of a connected corridor real-time data-driven traffic digital twin simulation model," *J. Transp. Eng. A, Syst.*, vol. 147, no. 12, Dec. 2021, Art. no. 04021096.
 - [27] D. Yang, H. R. Karimi, O. Kaynak, and S. Yin, "Developments of digital twin technologies in industrial, smart city and healthcare sectors: A survey," *Complex Eng. Syst.*, vol. 1, no. 1, p. 3, 2021.
 - [28] P. Novák, J. Vyskočil, and B. Wally, "The digital twin as a core component for industry 4.0 smart production planning," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 10803–10809, 2020.
 - [29] J. Zhang, T. Deng, H. Jiang, H. Chen, S. Qin, and G. Ding, "Bi-level dynamic scheduling architecture based on service unit digital twin agents," *J. Manuf. Syst.*, vol. 60, pp. 59–79, Jul. 2021.
 - [30] V. Kupriyanovsky *et al.*, "The ontologies of cyber-physical systems of the national digital twin of Great Britain and BIM on the examples of smart cities, railways, and other projects," *Int. J. Open Inf. Technol.*, vol. 9, no. 3, pp. 91–130, 2021.
 - [31] M. Li, Z. Li, X. Huang, and T. Qu, "Blockchain-based digital twin sharing platform for reconfigurable socialized manufacturing resource integration," *Int. J. Prod. Econ.*, vol. 240, Oct. 2021, Art. no. 108223.
 - [32] H. Feng, F. Bai, and Y. Xu, "Identification of critical roads in urban transportation network based on GPS trajectory data," *Phys. A, Stat. Mech. Appl.*, vol. 535, Dec. 2019, Art. no. 122337.
 - [33] E. Barmounakis and N. Geroliminis, "On the new era of urban traffic monitoring with massive drone data: The pNEUMA large-scale field experiment," *Transp. Res. C, Emerg. Technol.*, vol. 111, pp. 50–71, Feb. 2020.
 - [34] S. Wang, D. Yu, X. Ma, and X. Xing, "Analyzing urban traffic demand distribution and the correlation between traffic flow and the built environment based on detector data and POIs," *Eur. Transp. Res. Rev.*, vol. 10, no. 2, pp. 1–17, Jun. 2018.
 - [35] X. Wang *et al.*, "A city-wide real-time traffic management system: Enabling crowdsensing in social internet of vehicles," *IEEE Commun. Mag.*, vol. 56, no. 9, pp. 19–25, Sep. 2018.
 - [36] B. Ji *et al.*, "A survey of computational intelligence for 6G: Key technologies, applications and trends," *IEEE Trans. Ind. Informat.*, vol. 17, no. 10, pp. 7145–7154, Oct. 2021.
- Yang Liu** was born in Dalian, China, in 1987. He received the bachelor's degree in academy of arts and design from Tsinghua University in 2011 and the master's degree in business administration from the Dalian University of Technology in 2018. He is currently pursuing the Ph.D. degree with Dalian Maritime University. He also works with the Port and Logistics Hub Digitalization Laboratory, Dalian Maritime University. His current research interests include industrial design, digital twin, and maritime transportation infrastructure.
- Ang Yang** was born in Dalian, China, in 1987. He received the first bachelor's degree in information science, the second bachelor's degree (Hons.) in information technology, and the Ph.D. degree from the University of South Australian in 2013, 2014, and 2019, respectively.
- From 2015 to 2019, he worked as the Ph.D. Researcher with the Data to Decision Cooperative Research Centre (D2D CRC), Australia, in the fields of big data quality analysis. From 2014 to 2019, he also worked as an IT Consultant at Argen Company, Australia, for the development of ICT system. He is currently an Associate Professor with the School of Shipping Economics and Management, Dalian Maritime University, where he is also the Deputy Director of the Port and Logistics Hub Digitalization Laboratory. His research interests include big data quality, big data governance, information security, and maritime transportation.
- Dr. Yang was a recipient of the International President's Scholarship (IPS) and the Ph.D. Top-Up Scholarship in D2D CRC during his study.
- Qingcheng Zeng** was born in 1978. He received the bachelor's degree in international finance and the master's degree in management science and engineering from Harbin Engineering University in 2000 and 2003, respectively, and the Ph.D. degree in transport planning and management from Dalian Maritime University in 2008. He is currently a Professor with the School of Shipping Economics and Management, Dalian Maritime University. He is also the Dean of Strategy Planning and Academic Resources and the Director of the Port and Logistics Hub Digitalization Laboratory, Dalian Maritime University. His current research interests include maritime economics and logistics, port system simulation and optimization, and shipping operation management.
- Yuhui Sun** was born in Chengdu, China, in 1987. He received the bachelor's degree in pharmaceutical engineering from the East China University of Science and Technology in 2010 and the master's degree in chemical engineering from The University of Adelaide in 2013. He is currently pursuing the Ph.D. degree with the University of South Australia. Since 2014, he has been the Project Manager of Argen Company, Australia, and the Neusoft Education Group, China. His current research interests include building information modeling, digital twin, and maritime transportation infrastructure.
- Jing Gao** was born in 1980. He received the double bachelor's degree in computer science/management and the Ph.D. degree from the University of South Australia in 2001 and 2004, respectively, where he received First Class Honors in 2002. He is currently an Associate Professor in STEM with the University of South Australia. His research interests include higher education, management, engineering, information systems, and computer science. He is also an Australia Industry leading expert, specializing in design and implement enterprise information architecture, AI-driven corporate solutions (education technology and Industry 4.0) and data governance programs supporting both the public and private sector nationally and internationally. He have ongoing engagement to provide services to the top-level national and international organizations, such as BHP, Rio Tinto, Royal Australian Navy, Australian Taxation office, Department of Human Services, SA State Government, NSW State Government, Ministry of Education Singapore, and many Australian universities.
- Zhihan Lv** (Senior Member, IEEE) is with the School of Data Science and Software Engineering, Qingdao University, China.